

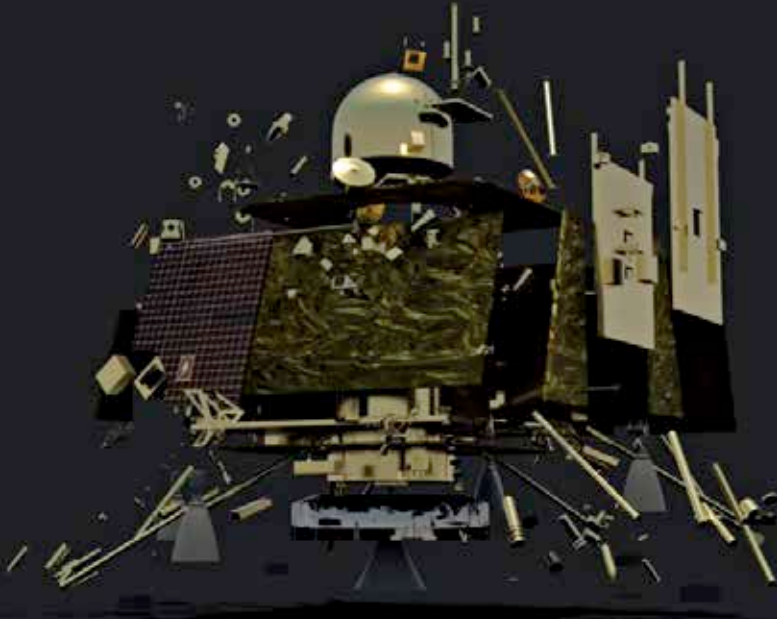
Dr Artificial Intelligence?

Studies indicate that AI is on par with human medical experts when it comes to interpreting medical images. <https://digit.in/oct19-51>



Water on super-Earth

Water vapour has been detected on super-Earth, K2-18 b's atmosphere, which has Earth-like temperatures. <https://digit.in/oct19-52>



chance of Vikram ending up in an area of communications shadow, because of the local terrain. Ahead of the launch, K Sivan described the tricky operation, "these fifteen minutes are going to be the most terrifying for all the people who belong to ISRO, probably for the scientists and the people in India also. This is going to be the most difficult task ever undertaken by ISRO in a mission". During the entire fifteen minutes, the thrusters had to be perpendicular to the surface of the Moon. The ISRO scientists also had to compare the images from the lander to previous images to make sure that the exact landing site did not have an inclination of more than 12 degrees. If the inclination was more,

CHANDRAYAAN 2

What went wrong

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ommunication link was lost with the Vikram lander in the last few nail biting minutes of the

controlled descent. ISRO had successfully executed most of the soft landing near the south pole of the Moon with the Vikram lander, as part of the Chandrayaan 2 mission, but the spacecraft appeared to veer off course just towards the end. A period of confusion followed, where ISRO personnel indicated that they were not receiving any communications from the spacecraft. Officially, ISRO Chairperson K Sivan said, "Vikram lander descent was as planned, and normal performance was observed up to an altitude of 2.1 kilometers. Subsequently, the communications from the lander to ground station was lost. The data is being analysed." There are also some

indications from the trajectory that there may have been an attitude correction manoeuvre.

The landing was the trickiest part of the mission, and the most technically-sophisticated operation undertaken by ISRO in its history. One of the primary objectives of the mission was to demonstrate the capability of being able to execute a soft landing.

FIFTEEN TERRIFYING MINUTES

The site of the landing was a high plane between the Simpeliuss N and Manzinus C craters, at a latitude of 70.9° South and a longitude of 22.7° East. An alternative landing site was 67.7° South 18.4° West, and perhaps things could have played out differently if this site were chosen. Vikram had a touchdown velocity of 2 meters per second. Considering the velocity, there was a chance that the lander could have crashed into the Moon, topple over or be unable to communicate, if things didn't go exactly as planned. The landing site also had to be carefully chosen to eliminate any

then Vikram would have toppled after the landing. The lander went through thorough drop testing at a specially designed test facility in Chitradurga, that simulated the terrain of the Moon. This was the first time that ISRO had undertaken the in-house development of a lander and a rover, including all the complicated systems necessary to execute a soft landing. These included the four independent thrusters between the legs for steering and stabilisation, as well as a fifth thruster below the lander for additional stabilisation.

The lander was also loaded with sensors placed all around its body to aid with the navigation and soft landing. These included a position detection camera, a horizontal velocity detection camera, a hazard detection and avoidance camera, two k-band altimeters and a laser altimeter.

The descent operations started from an altitude of 30 kilometers. The entire landing sequence was autonomous, without any command and control from the ground stations. The algorithms that were necessary to